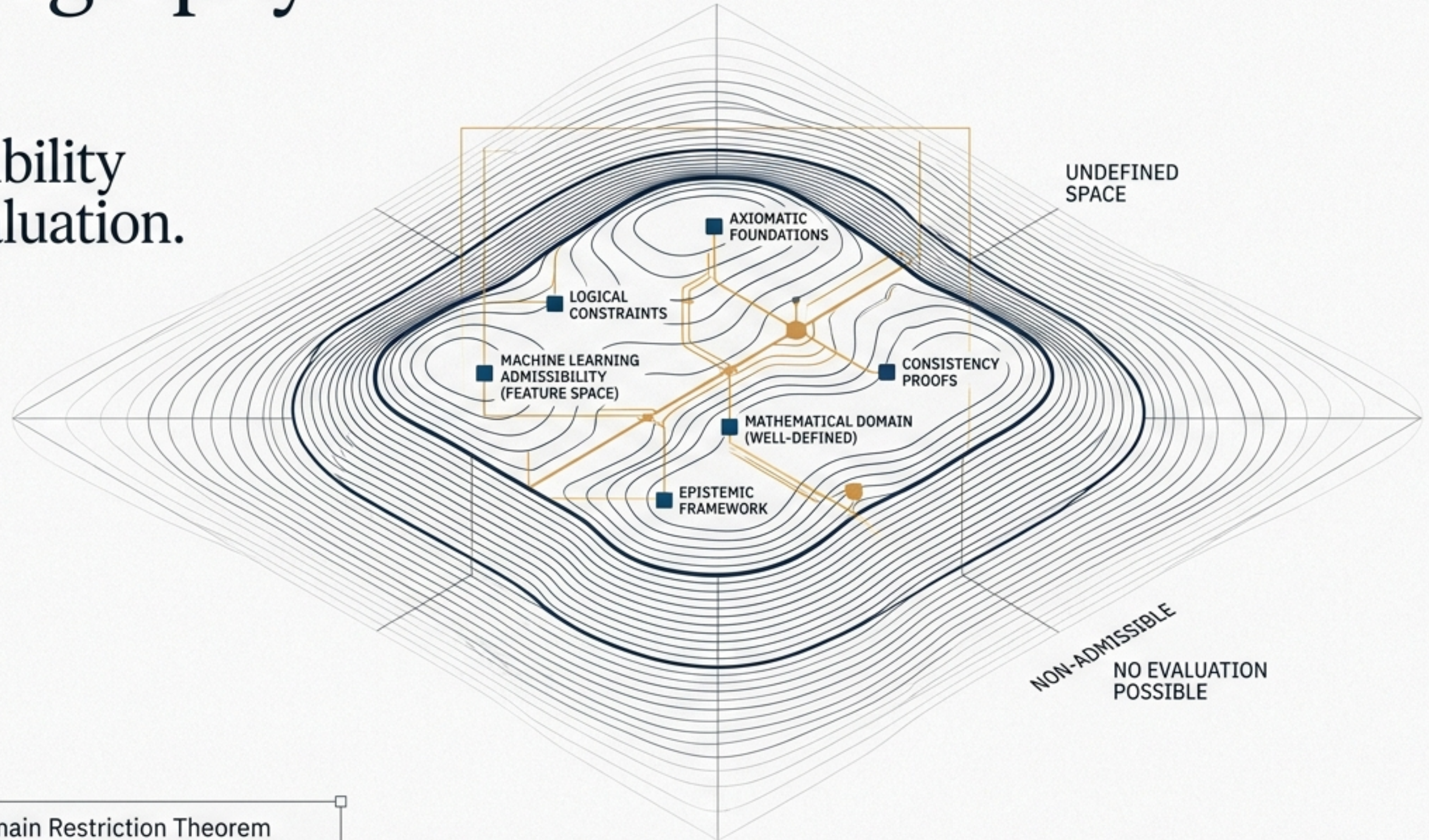


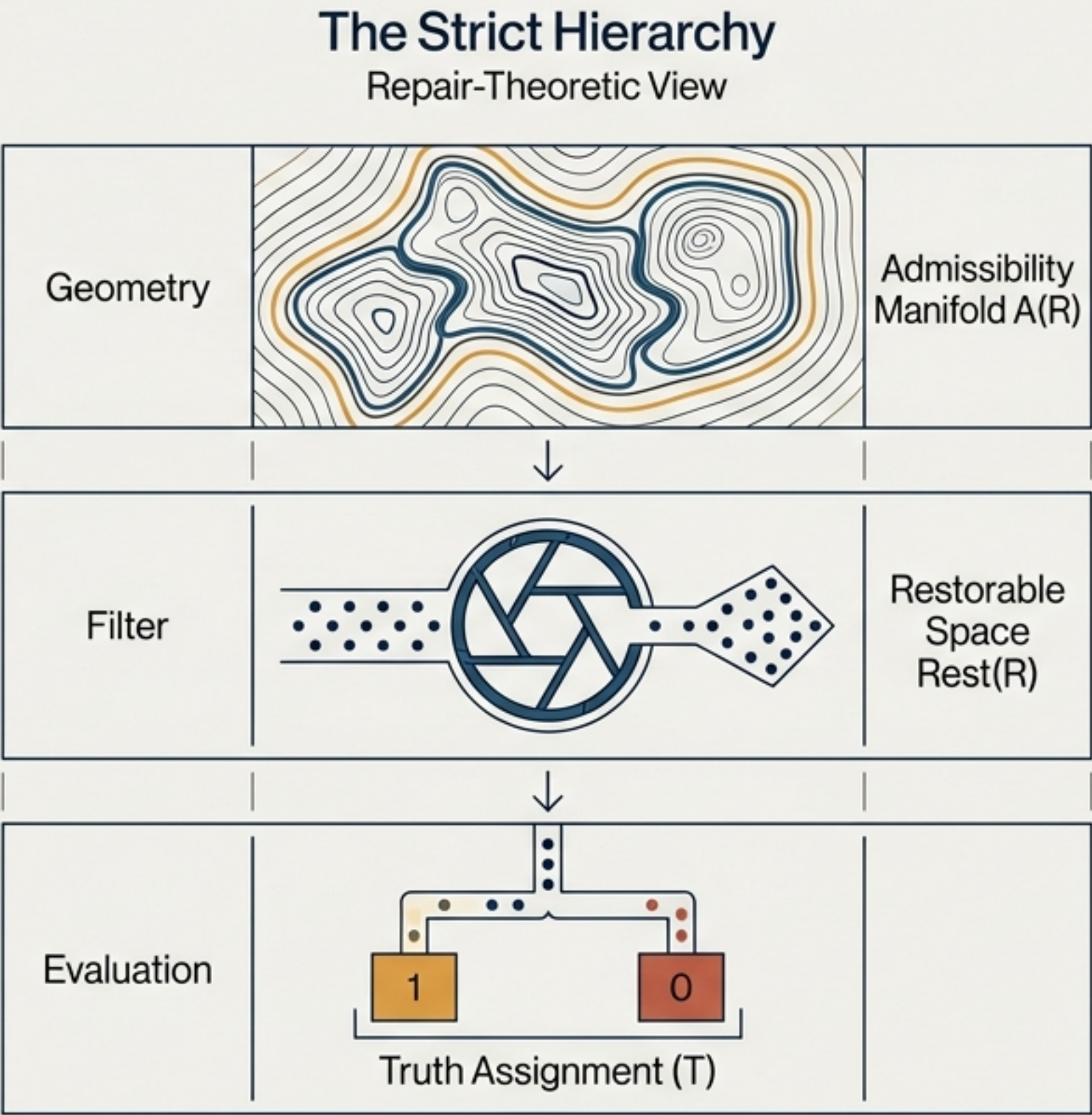
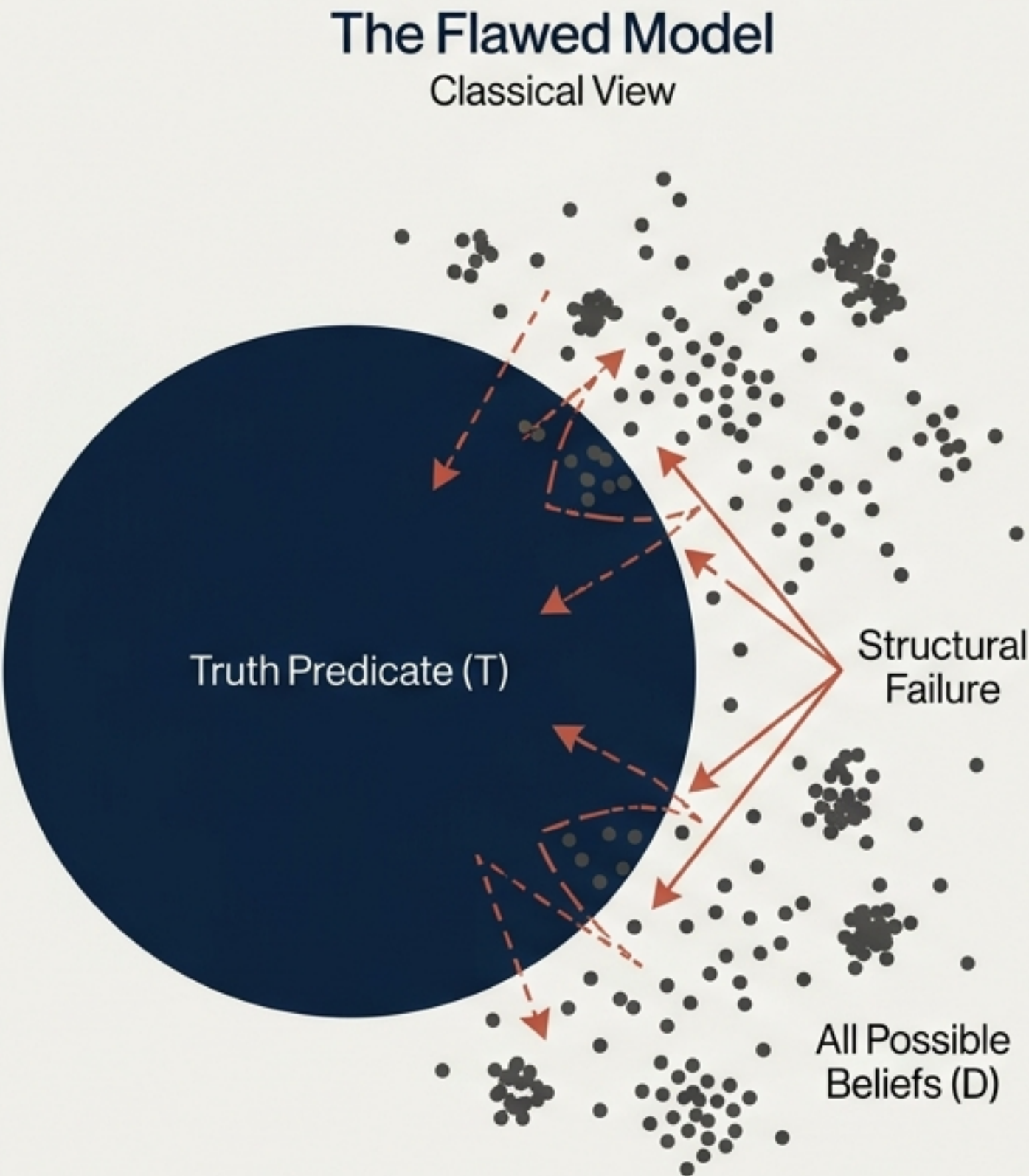
The Topography of Truth

Why Admissibility
Precedes Evaluation.



A structural proof of the Domain Restriction Theorem
across mathematics, machine learning, and epistemology.

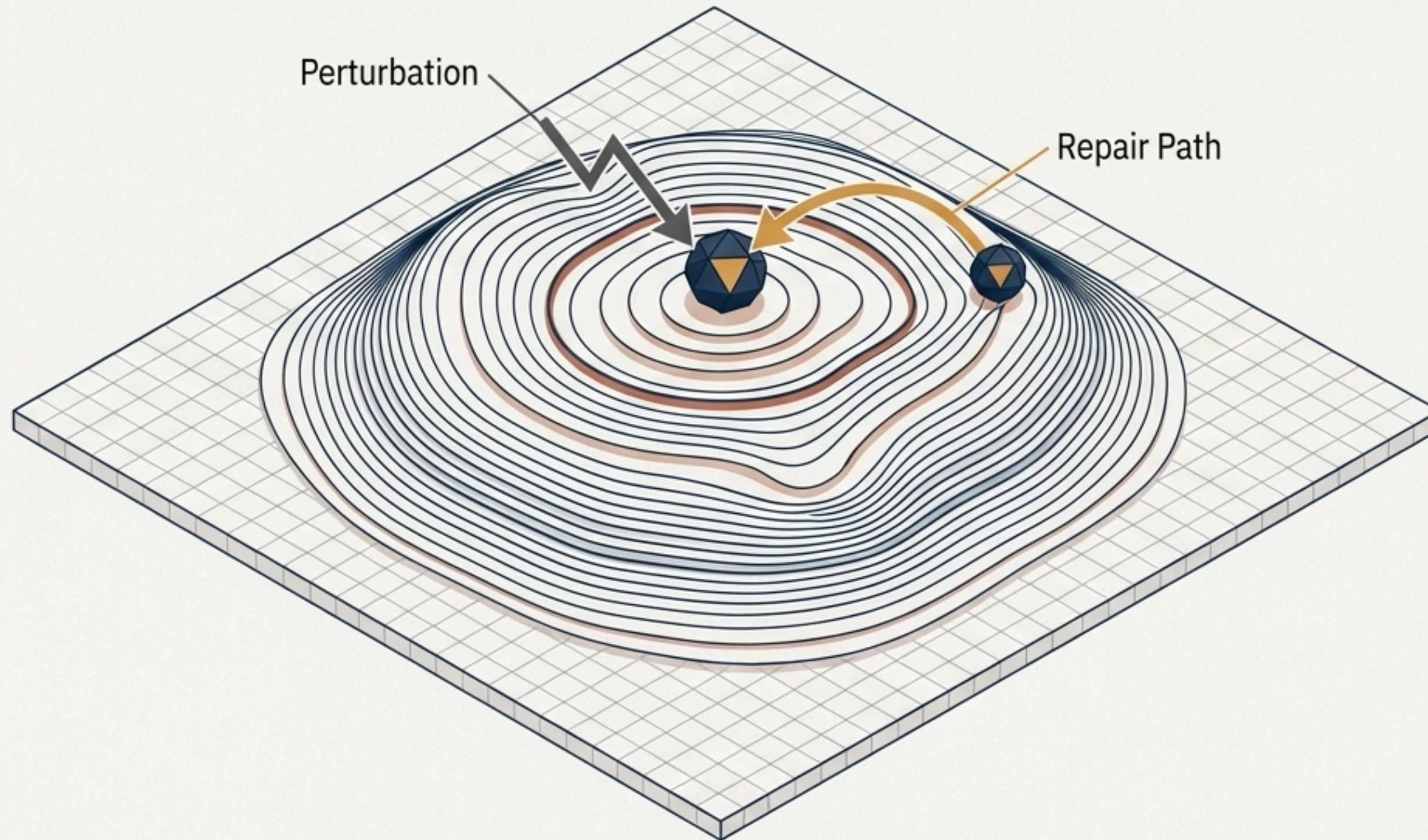
Standard epistemology asks which beliefs are true. The Domain Restriction Theorem proves this order is backward: an observer must first establish which distinctions it can maintain, and only then assign truth values.



The Topography of Evaluation

D: The Full Distinction Space

A(R): The Admissibility Manifold



R (Repair Repertoire)

The observer's finite toolkit of continuous maps capable of restoring state after perturbation.

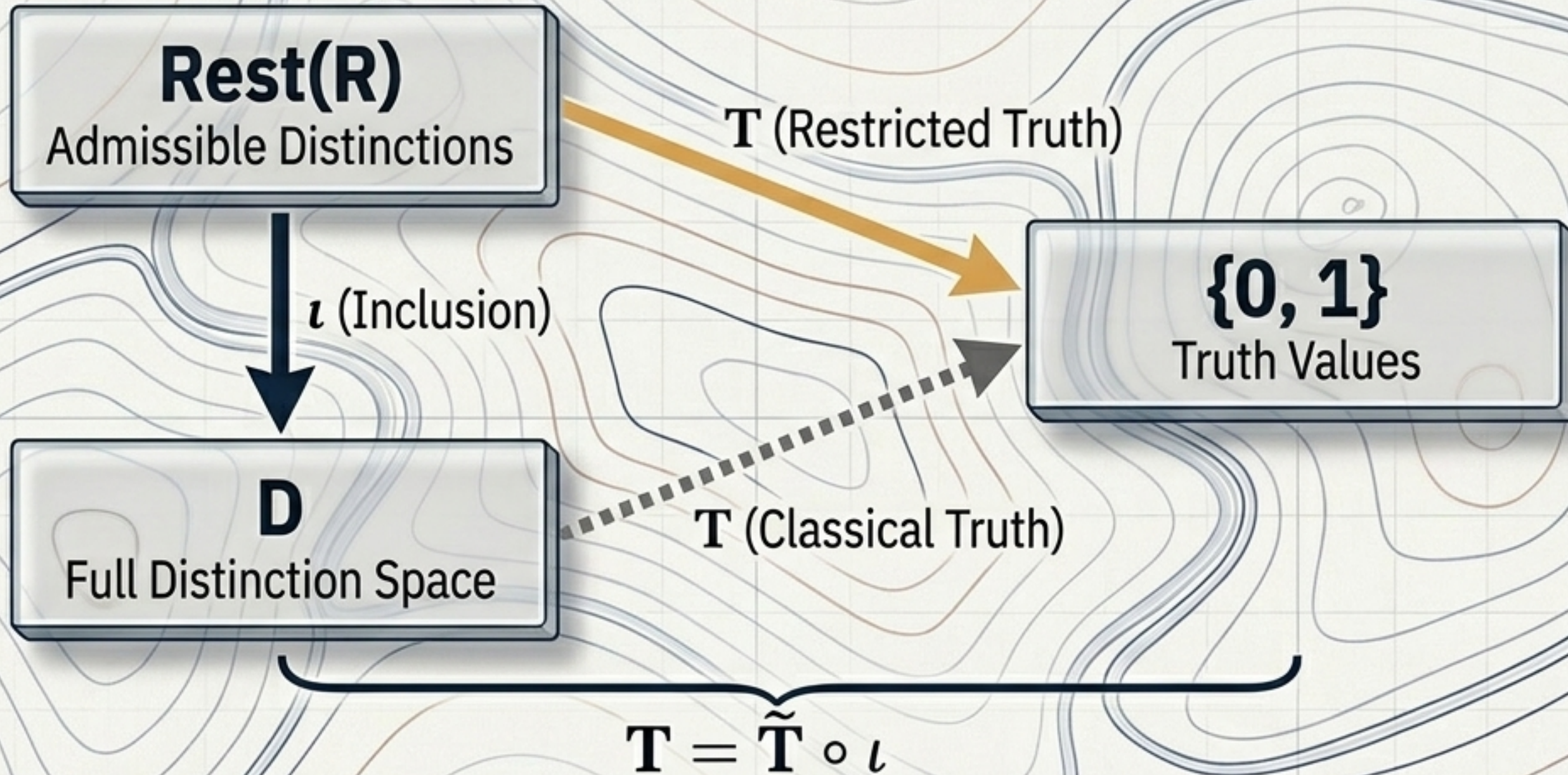
Rest(R) (Restorable Space)

The exact set of distinctions for which finite-cost repair paths exist.

The Geometric Reality

What the observer cannot repair, the observer cannot evaluate. It is simply not part of their world.

The Factorization Map

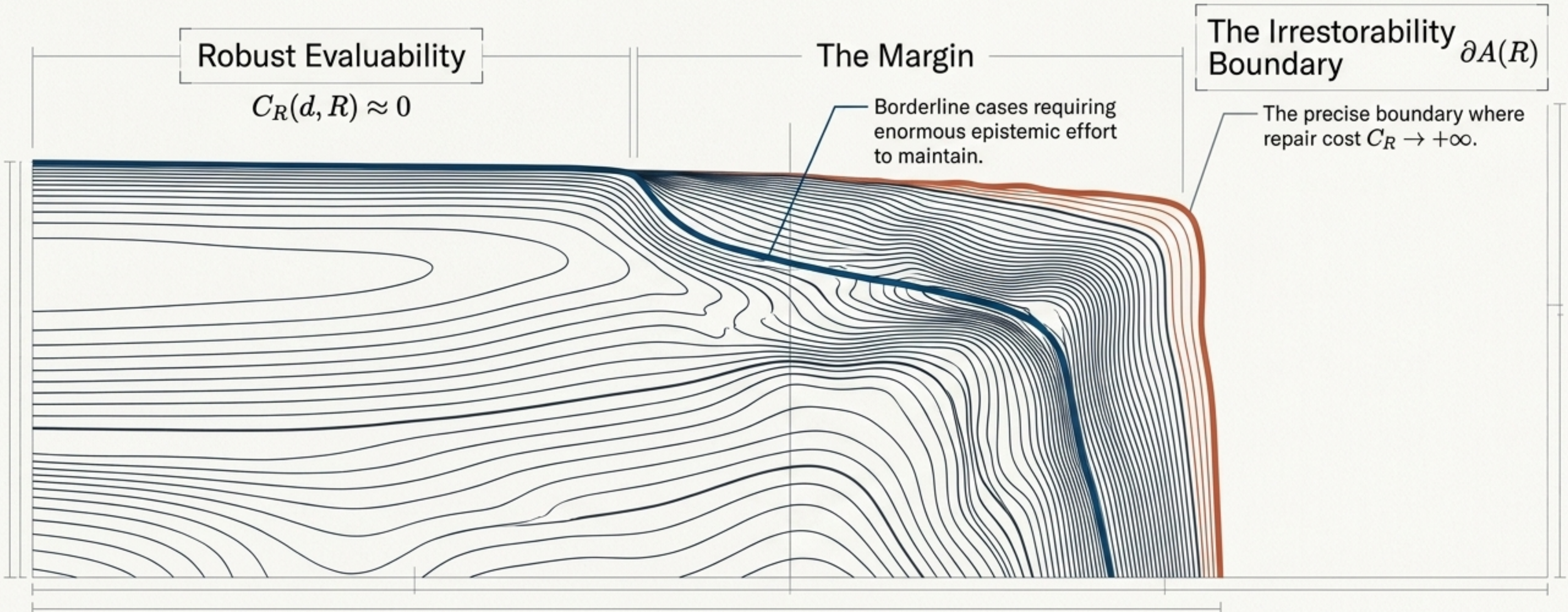


The Domain Restriction Theorem

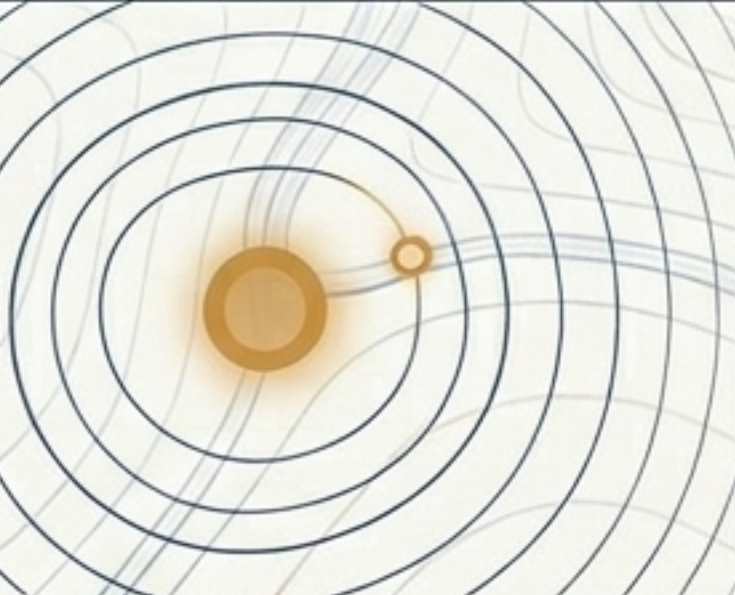
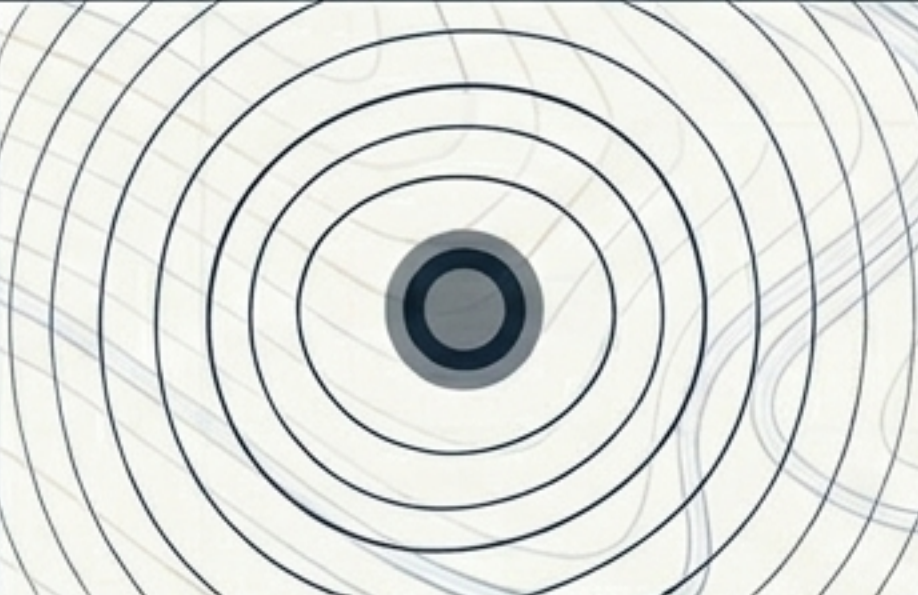
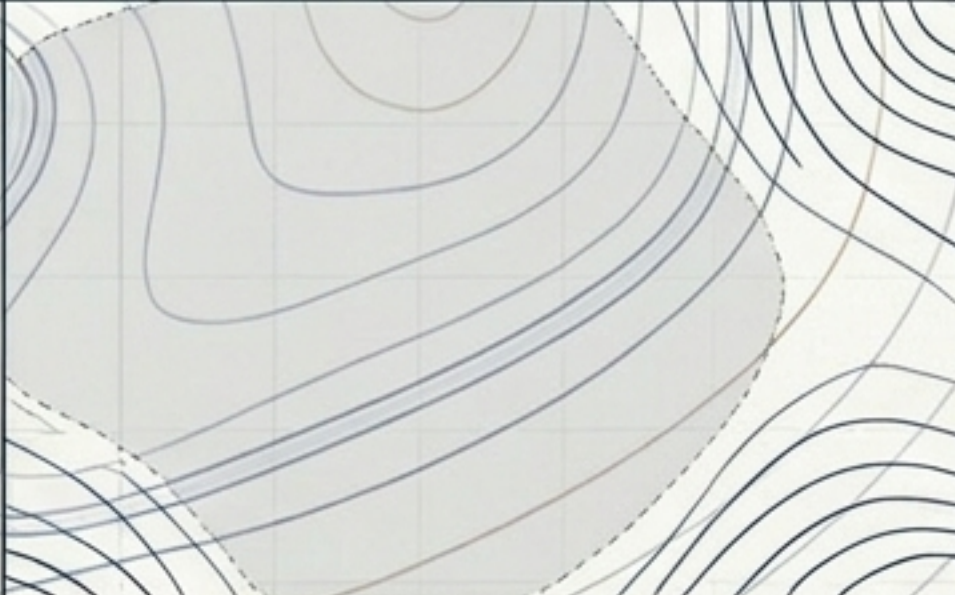
Every well-defined truth predicate literally factors through the inclusion map. The Truth Predicate cannot bypass the Admissibility Manifold. $\text{dom}(T) \subseteq \text{Rest}(R)$

The Evaluability Terrain

The boundary of truth inherits the geometric structure of the admissibility manifold. Beyond this cliff, truth does not fail—it simply ceases to operate.



Three-Valued Semantics

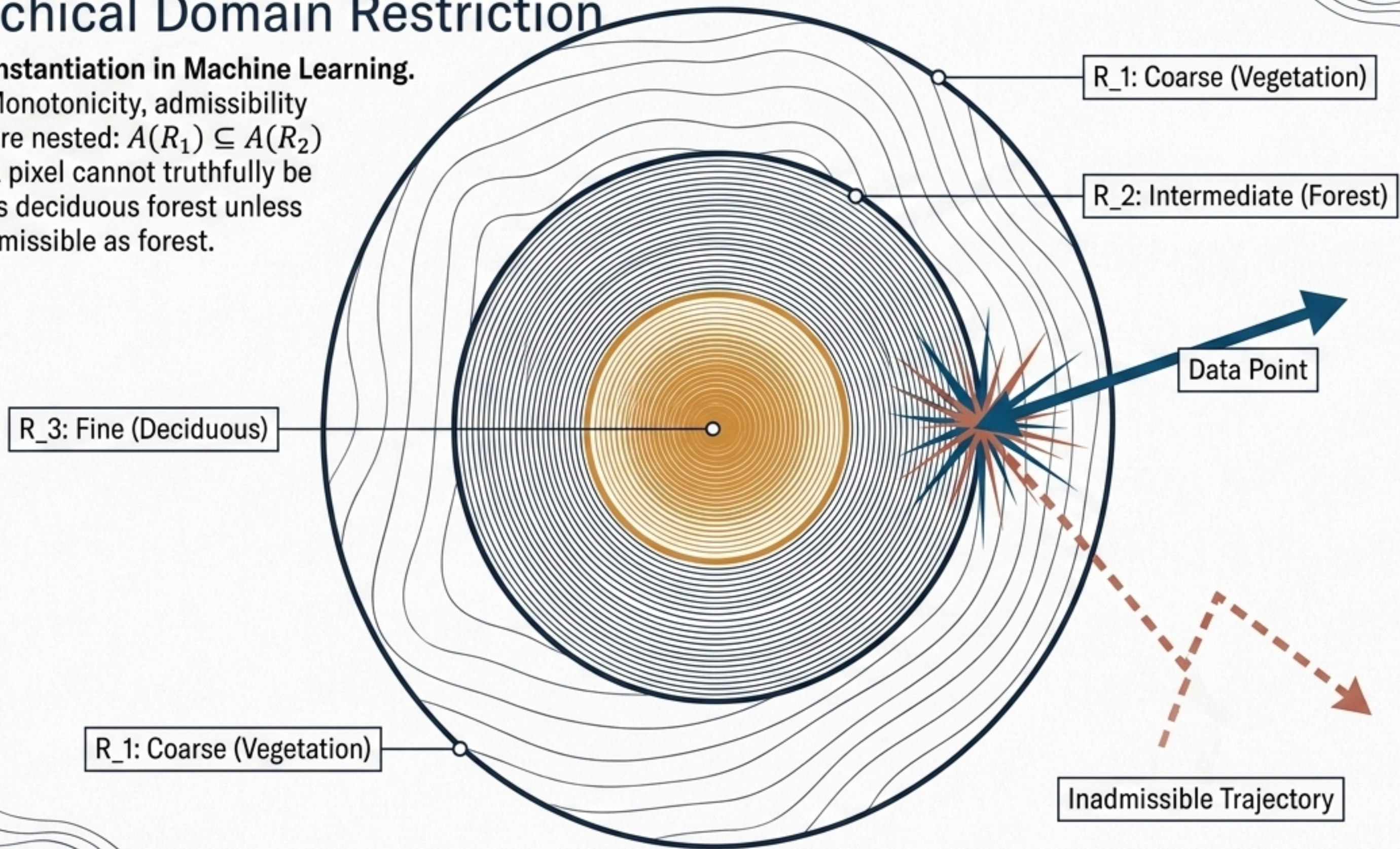
	Truth (1)	Falsity (0)	Unevaluability ($\perp$)
Mathematical Condition	$d \in \text{Rest}(R)$ AND condition holds.	$d \in \text{Rest}(R)$ AND condition fails.	$d \notin \text{Rest}(R)$.
The Meaning	The distinction is verified.	The distinction is discriminably absent.	Failure of precondition. The repair cost is infinite.
Visual Metaphor			

$\perp$ is not indeterminate or false by default. It is a strictly distinct value arising from geometric inadmissibility.

Hierarchical Domain Restriction

Empirical Instantiation in Machine Learning.

By Repair Monotonicity, admissibility manifolds are nested: $A(R_1) \subseteq A(R_2) \subseteq A(R_3)$. A pixel cannot truthfully be classified as deciduous forest unless it is first admissible as forest.



The Consensus Loss as Geometric Reality

State-of-the-art neural networks independently converged on Theorem 4.1. Consensus loss penalizes truth assignments made to distinctions that cannot be projected into coarser admissibility domains.

The ML Equation

SAHC Hierarchical Consensus Loss

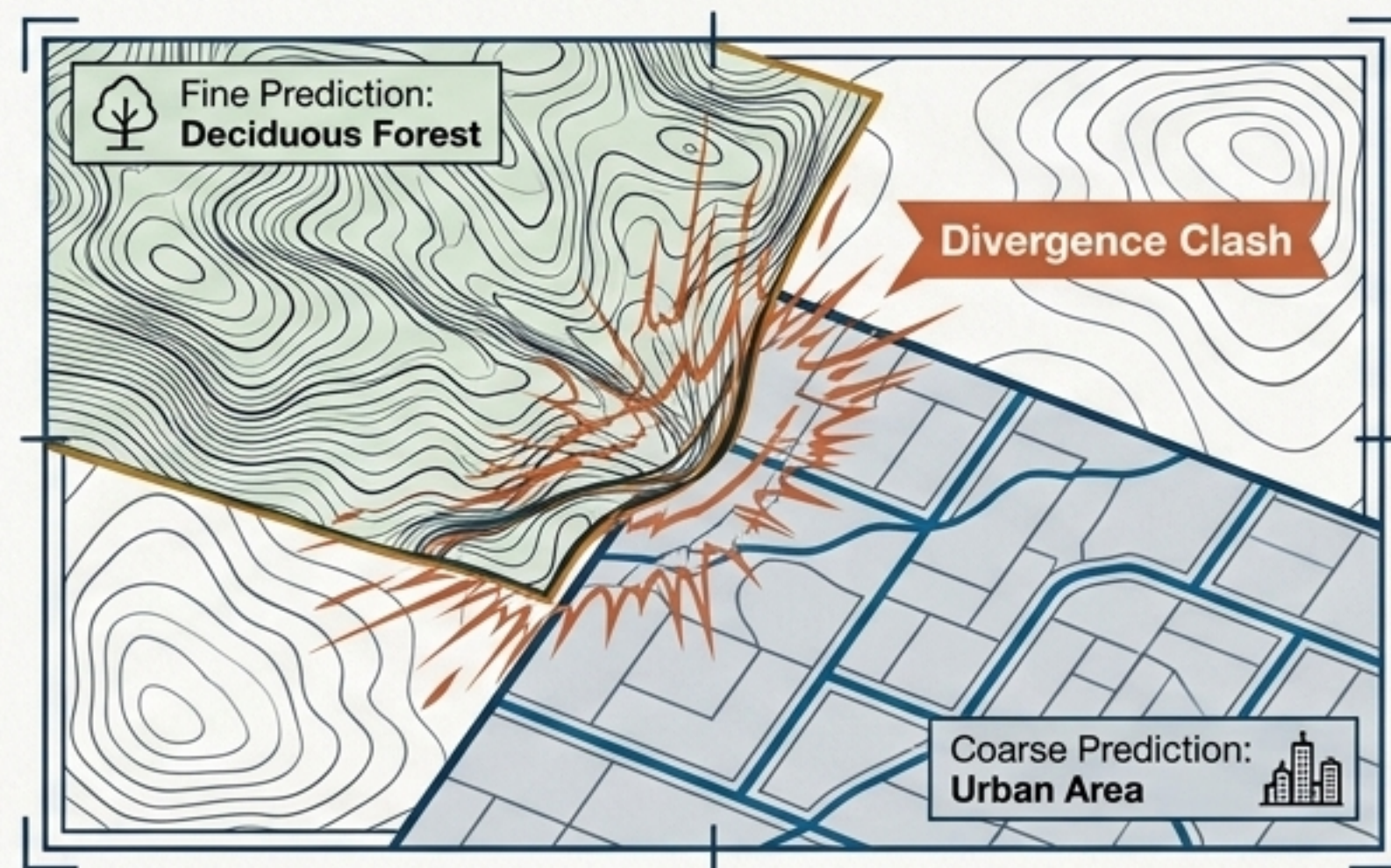
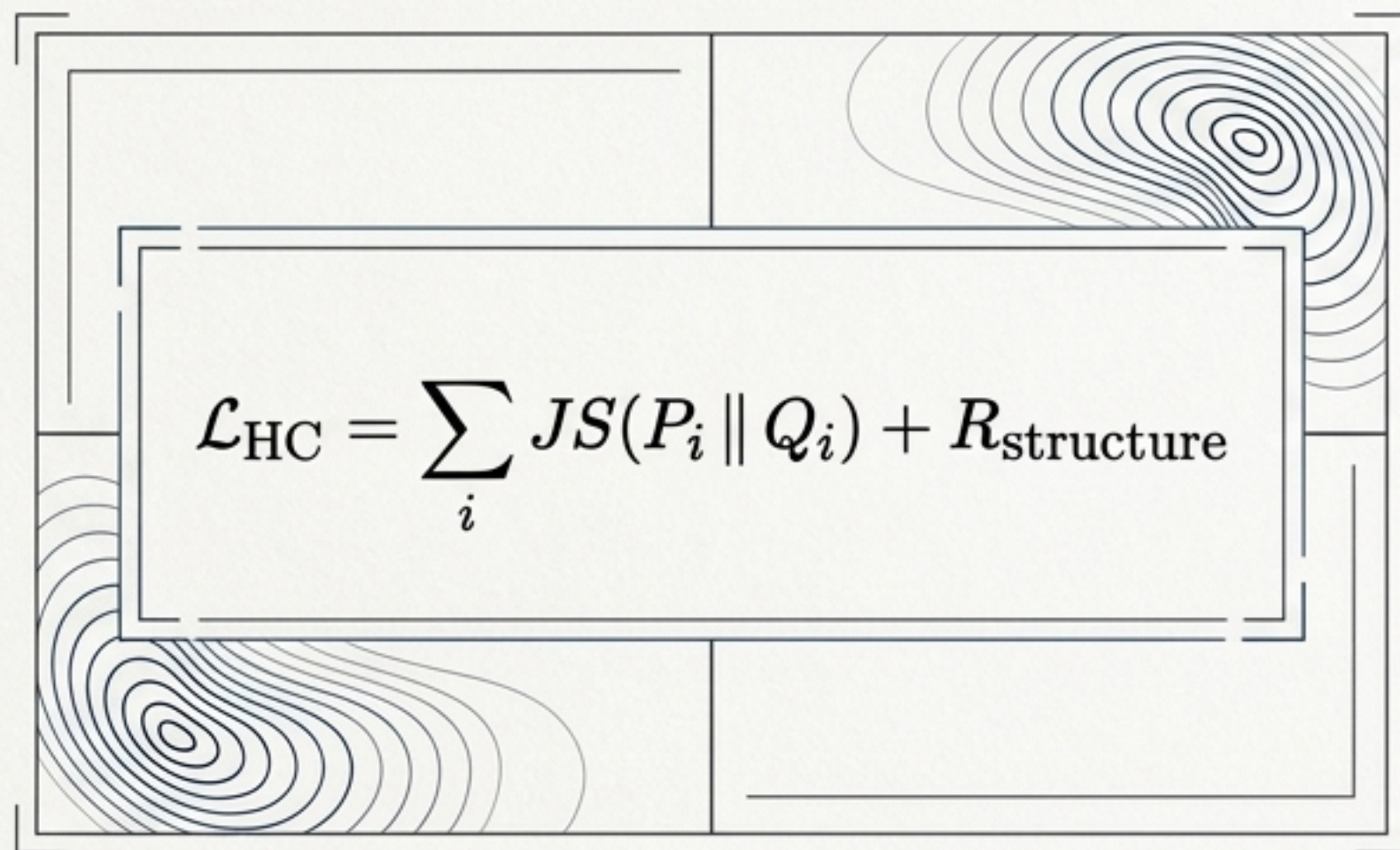
$\mathcal{L}_{HC}$ penalizing Jensen-Shannon divergence

$$\mathcal{L}_{HC} = \sum_i JS(P_i \parallel Q_i) + R_{\text{structure}}$$

The Geometric Translation

Domain Restriction Penalty

$$\mathcal{L}_{HC} \approx d(\tilde{T} \circ \iota, T)$$



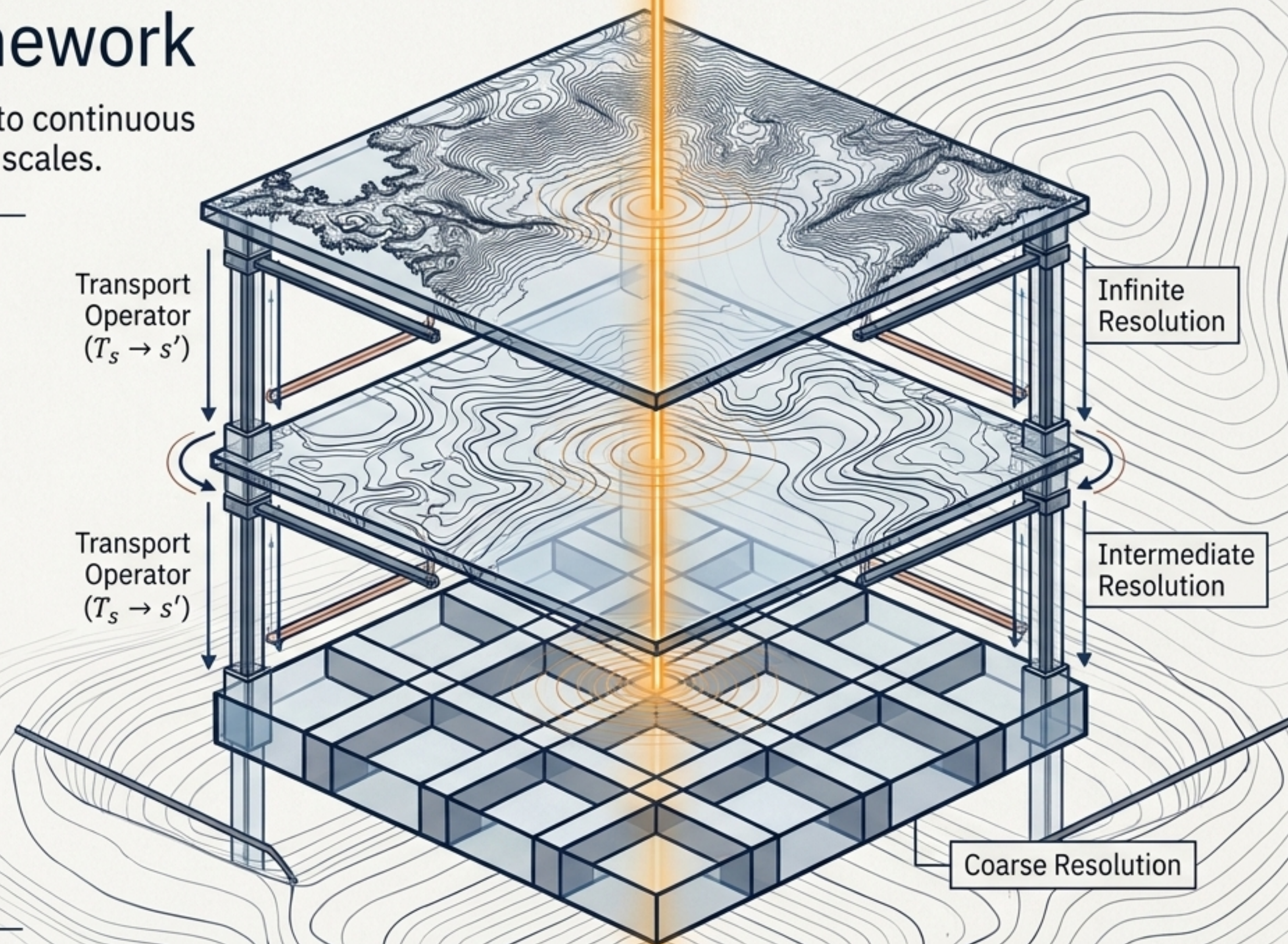
The TARTAN Framework

Generalizing the restriction theorem to continuous spatial, temporal, and computational scales.

Continuous
Resolution Tower
($s \in [0, \infty)$)

Zero coherence ($L_{\text{TARTAN}} = 0$)
implies **perfect** consistent
domain restriction.

Evaluability domains grow
strictly monotonically
with resolution.



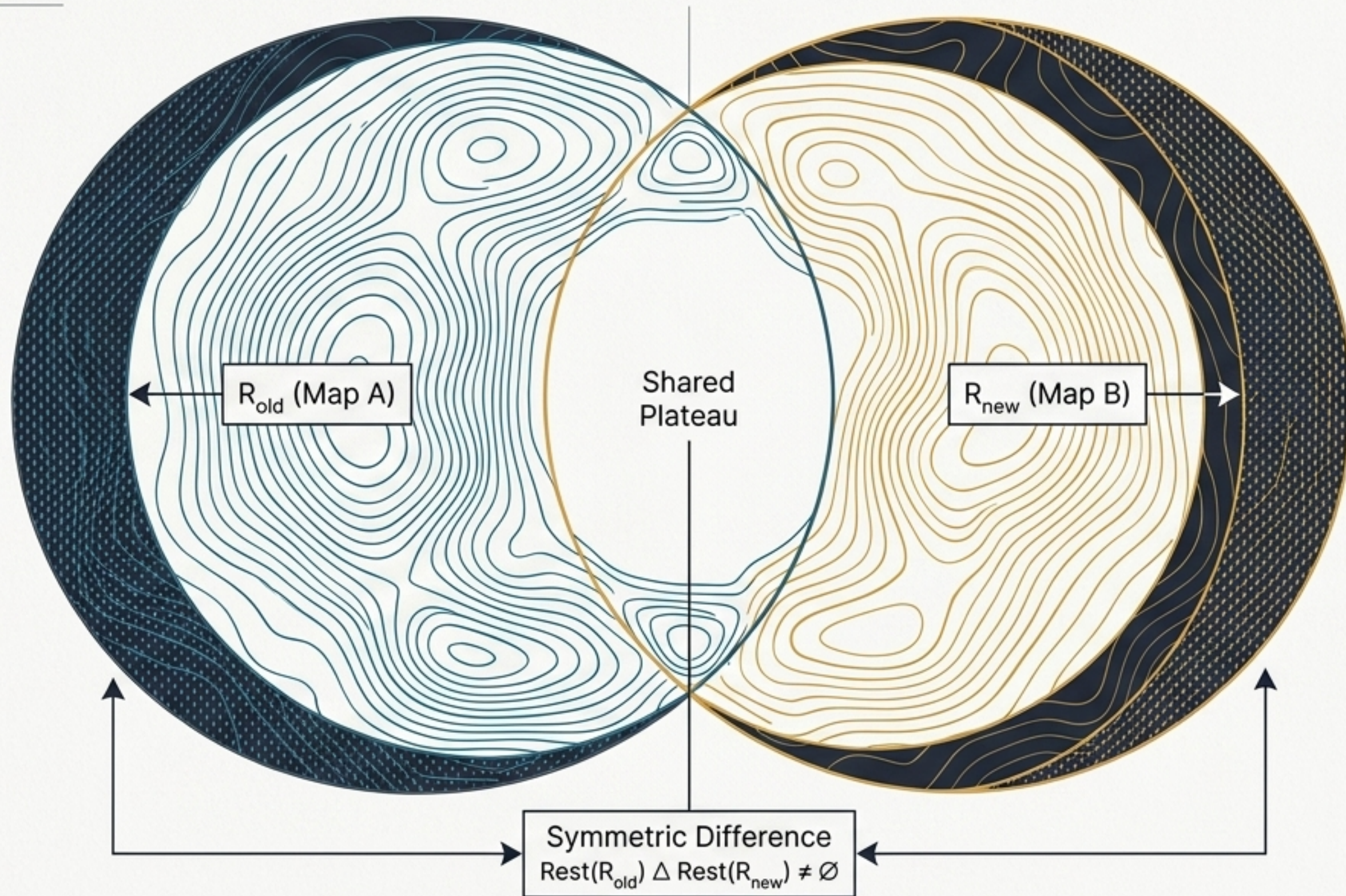
The Paradigm-Shift Synthesis

Anomalies in disparate fields are independently discovered instances of identical domain restriction.

Discipline	The Observer	Repair Repertoire (R)	Unevaluable State ($\perp$)
Machine Learning	A trained classifier	The training distribution	Out-of-Distribution (OOD) Failure
Law	The Court	Rules of Evidence / Procedure	Legally Inadmissible
Linguistics	A Language Speaker	Syntactic / Semantic Grammar	Ungrammatical (No proposition)
Bayesian Inference	A Rational Agent	Coherent Prior Support	Incoherent (Outside the prior)
Science	A Scientific Community	A Paradigm's Methodologies	Kuhnian Incommensurability

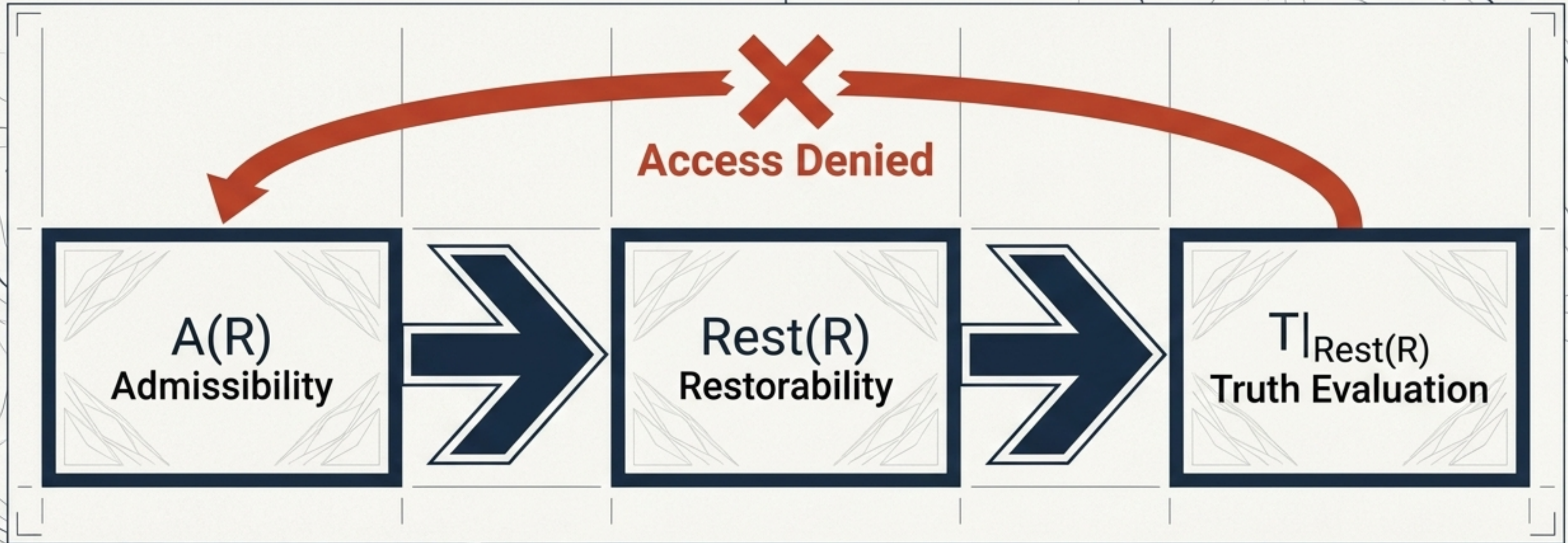
Kuhn Without Relativism

Truth does not change, but domains of evaluability differ. Claims in the symmetric difference transition from evaluable to unevaluable based on the shifting repair repertoire.

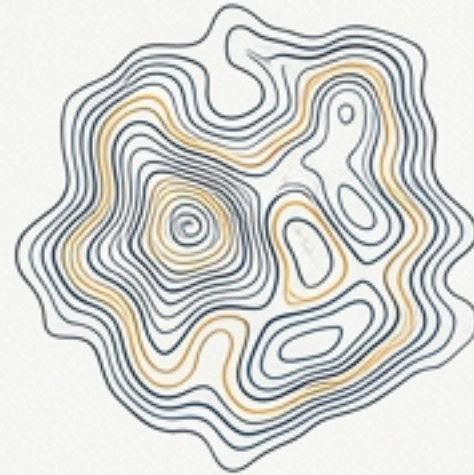


The Ordering Does Not Reverse

Truth-evaluation operates exclusively on distinctions already within Rest(R). Assigning a 0 or a 1 does not create new repair operators. You cannot evaluate your way into a larger world.



Admissibility is not a property truth has or lacks.
It is the precondition for truth having anything to say.



Epistemology	Science & Learning	Language
The regress terminates not in foundational truth, but in a repair repertoire.	Progress is not the accumulation of true beliefs, but the expansion of the boundary of evaluability.	The admissibility manifold is the formal, geometric home for the limits of one's world.